Why Are HPC Monitoring Signals Hard to Forecast?

Intrinsic Predictability Across Eight Telemetry Families of a Tier-0 Supercomputer

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Abstract

Operational telemetry from high-performance computing systems resists forecasting to a degree that is widely reported anecdotally and rarely explained. The usual response is to increase model capacity. We test the opposite position: that the difficulty is largely *intrinsic to the signals*, differs systematically between signal families, and can be quantified without fitting any forecaster. Half of that position survives our own test and half does not, and we report both.

Using the public M100 ExaData release from CINECA's Marconi100 Tier-0 supercomputer, we characterise telemetry from eight heterogeneous plugin families — out-of-band sensors, operating-system counters, scheduler aggregates, alarm states, facility power, room-level electrical measurement, cooling-plant control registers and air handlers, and exogenous weather — collected from the same machine over the same 31 months. (A ninth family, per-job submission records, is collected but is event-structured rather than a time series, and is excluded; OI-8.) The families are held fixed in system, workload and period while signal type varies. For each series we compute model-free predictability measures spanning entropy, periodicity, long memory, non-stationarity, and quantisation, and separately measure the forecast skill achieved by a tiered suite ranging from naive baselines to gradient-boosted trees under a strictly causal rolling-origin protocol.

We report four findings. First, the monitoring infrastructure is itself non-stationary: of 573 distinct (plugin, metric) pairs observed across the campaign, only 104 are present in every month, and those are exactly the out-of-band sensor set. Second, a large share of what is monitored carries no forecastable information at all — the fraction of series that never change over a month ranges from 64.8% of alarm-state signals to 8.4% of out-of-band sensors — so any benchmark that averages over such series reports the composition of its sample as much as the skill of its models. Third, model capacity does not help: measured against a same-step naive baseline, median skill by tier at a one-hour horizon runs 0.115 for naive forecasters, -0.064 for statistical ones, -0.327 for neural networks and -0.613 for gradient-boosted trees, and beyond three hours every tier is worse than repeating the last value.

Fourth, and contrary to our hypothesis, the measures transfer across signal families in *ordering* but not in *level*. A model trained on every family except one ranks the held-out family's series correctly — median within-family Spearman +0.50, significant for 9 of 15 families and reaching +0.83 — while assigning the wrong absolute skill, which is what drives the raw R^2 to a mean of -0.19. Correcting the level bias moves the mean to -0.01 and makes 10 of 15 families positive. A random-group null settles what this means: the same protocol run with arbitrary groups of matched size transfers easily (median R^2 +0.33 across draws, every group positive, $p < 0.001$ against the observed value over 100 draws), so the difficulty is specific to crossing family boundaries rather than a limitation of the protocol. Practically, the profile can rank which signals in an unfamiliar family will be harder, but not say how hard.

1 Introduction

Operators of large computing facilities increasingly collect holistic telemetry, and increasingly want to *predict* it rather than merely react to it: thermal headroom, power capping, and proactive maintenance are all forecasting problems. Reported accuracy on this class of signal is nonetheless poor relative to standard time-series benchmarks, and the literature generally treats that as a modelling deficiency to be fixed with a stronger model.

We invert the question. Instead of asking which model does best, we ask how much predictability is present at all, measured independently of any forecaster, and whether that measurement predicts the skill a forecaster subsequently achieves. If it does, "HPC telemetry is hard to forecast" becomes a quantity with named causes rather than folklore.

Our contributions are: (i) a quantitative census of signal heterogeneity across a 31-month Tier-0 monitoring campaign, including the instability of the monitoring system itself; (ii) a model-free predictability profile for each of 1402 series drawn from eight plugin families; (iii) a strictly causal, scale-free forecast benchmark over four model tiers on the same series; and (iv) a test of whether the first predicts the second, cross-validated by holding out entire signal families.

The headline result is negative, and we report it as such: the measures do not transfer across families. We regard this as the more useful outcome than a confirmation would have been, because the experiment was constructed so that the hypothesis could fail, and identifying *which* part fails narrows what a future predictability theory has to explain.

2 Background

2.1 Marconi100 and the ExaData campaign

Marconi100 was a Tier-0 system at CINECA comprising 980 IBM Power AC922 nodes, each with two 16-core POWER9 processors and four NVIDIA V100 accelerators. The ExaData campaign published holistic telemetry for the machine from March 2020 to September 2022, collected through the ExaMon framework and released as Hive-partitioned Parquet, one archive per month.

We catalogue every partition of all 31 archives — 10504 partitions holding roughly 55 billion rows — and extract the subset the study requires. Because node-level series of one metric are highly redundant, and because nodes sharing a rack, inlet air and workload are not independent samples (OI-9), analysis runs on a fixed panel of 96 *primary* entities — nodes, cooling devices or PLC panels — whose membership is a stable hash of the entity label. The hash sees only the label, so panel membership is identical across every metric and month and cross-metric comparisons land on the same physical machines. Families keyed by a secondary tag yield proportionally more series than the cap suggests: Nagios identifies a series by node *and* check description and reaches 1484 distinct series in a single metric-month, where the 96 bound applies to the node dimension alone.

2.2 The nine plugin families

The value of this dataset for the present question is that it holds the system fixed while varying the signal. Table 1 summarises the families and the mechanism that generates each.

3 Related work

Four bodies of work bear on this study. Claims below were checked against the source papers rather than their abstracts; the four we could obtain lawfully are read and verified, and the remainder are cited only for definitions we implement (OI-10).

Table 1: Telemetry families and their generating mechanisms. Nominal cadences are as documented by the dataset; measured cadences differ substantially for some families (§5.1).

Family	Measures	Nominal cadence	Generating mechanism
IPMI	BMC sensors: temperature, power, fans	20 s	thermal diffusion, quantised
Ganglia	OS and GPU counters	20–300 s, irregular	software instrumentation
SLURM	cluster scheduler aggregates	5–10 s	user submission behaviour
Nagios	service and hardware alarm states	15 min	discrete state machine
Job table	per-job submission records	event-driven	not a time series (OI-8)
Schneider	liquid-cooling PLC registers	20 s	closed-loop control
Logics	room power, PUE, chillers	—	facility aggregation
Vertiv	air-handling units	10 s	closed-loop control
Weather	outdoor conditions	10 min	exogenous

The dataset. M100 ExaData [Borghesi et al., 2023] publishes 49.9 TB of holistic telemetry — management, workload, facility and infrastructure — from Marconi100 across two and a half years. Two of its derived releases matter here. The unsupervised anomaly-detection set aggregates *IPMI* metrics into 15-minute periods with average, standard deviation, minimum and maximum per window, plus a Nagios-derived label; our common analysis grid matches that choice, though we apply it to all nine families rather than IPMI alone. The thermal-hazard set reduces sampling from 20 seconds to 10 minutes, and the authors state the trade explicitly: that rate "will preserve room- and node-level thermal transient but discard computing-component-level ones". That is a direct statement of what aggregation costs, and it applies to our results as much as to theirs (OI-4).

What this telemetry is normally used for. The dominant application is anomaly *detection* rather than forecasting. Borghesi et al. [2019] train per-node autoencoders on HPC monitoring data and report detecting previously unseen anomalies at accuracies between 88% and 96%. Detection asks whether the present sample is unusual given a learned model of normal behaviour; forecasting asks what the next sample will be. The former can succeed on signals whose future values are not predictable at all, which is one reason the difficulty we measure has attracted little attention.

Feature-based characterisation of time series. Our method follows the tradition of describing a series by a fixed feature vector rather than by a fitted model. Lubba et al. [2019] reduce 4791 candidate features to 22 by selecting for classification performance and minimal redundancy across 93 datasets, achieving a roughly 1000-fold reduction in computation for an average accuracy loss of 7%, and covering linear and non-linear autocorrelation, successive differences, value distributions and outliers, and fluctuation scaling. Our profile is smaller and chosen for interpretability by mechanism rather than by classification performance, but the premise — that a compact, model-free description captures much of what matters about a series — is theirs.

Capacity and forecast accuracy. Our finding that heavier models do worse has a direct precedent outside HPC. Zeng et al. [2023] introduce one-layer linear models and report that they outperform sophisticated Transformer-based long-term forecasters "in all cases, and often by a large margin" across nine real-world datasets, arguing that permutation-invariant self-attention discards temporal ordering. Our result is the same shape in a different domain and at the opposite end of the complexity range: on this telemetry a persistence forecast beats gradient-boosted trees and neural networks at every horizon tested. Where they compare a linear model against Transformers, we find the useful baseline is simpler still.

What we did not verify. MASE [Hyndman and Koehler, 2006], permutation entropy [Bandt and Pompe, 2002], detrended fluctuation analysis [Peng et al., 1994] and the Theta method [Assimakopoulos and Nikolopoulos, 2000] are cited for definitions we implement and test against known-answer series. Their full texts are paywalled and were not read; no claim is made here about their findings beyond the definitions, and we attribute no result to them. Three of the four works we did read were obtained as preprints rather than the versions of record we cite, which we note as a limitation of the audit rather than of the works.

4 Method

4.1 Campaign census

Tar members are read header-only, so cataloguing all 31 archives costs seconds and no disk. For each extracted partition we then compute a per-series census: one row per (plugin, metric, entity, month) recording sample count, observed cadence, cadence jitter, gap fraction, coverage, missingness, value range, distinct-value count and quantisation step. The census is deliberately descriptive — it contains no predictability measure and no model — so that a later finding of between-family differences cannot be an artefact of having looked first.

4.2 The entity panel

Node-level series of one metric are highly redundant — nodes share racks, inlet air, cooling zones, and workload — and are therefore not independent samples (OI-9). Analysis runs on a fixed panel of 96 entities whose membership is a stable hash of the entity label under the project seed. Because the hash sees only the label, panel membership is identical across every metric and every month, so cross-metric comparisons are made on the same physical machines rather than on a fresh random slice.

4.3 A-priori signal taxonomy

Labels are assigned from the plugin and metric *name* alone, before any predictability is measured, so that a finding of between-family differences cannot be a relabelling of the result it is meant to explain.

4.4 Model-free predictability measures

Each series receives a profile of 21 measures grouped by the mechanism it probes: entropy and complexity (permutation entropy [Bandt and Pompe, 2002], spectral entropy, sample entropy [Richman and Moorman, 2000], Lempel–Ziv complexity, gzip compression ratio); periodicity (dominant-period strength, daily seasonal strength); memory (autocorrelation at lag one and one day, detrended fluctuation exponent [Peng et al., 1994]); stationarity (block-mean drift ratio, variance ratio, Augmented Dickey–Fuller [Dickey and Fuller, 1979] and KPSS [Kwiatkowski et al., 1992] p-values, and a mean-shift changepoint rate obtained by binary segmentation [Truong et al., 2020]); and shape (jump fraction, repetition fraction, monotone fraction, reset rate, excess kurtosis of differences). Four instrument covariates — quantisation step, interval jitter, gap fraction and coverage — are carried separately so their contribution can be isolated from signal structure.

Two of these were added in response to an earlier failure of this study's own analysis rather than chosen a priori, and we flag the distinction. Monotone fraction and reset rate target counter-like signals, which a drift forecaster handles almost exactly and which the original profile could not distinguish from genuinely hard signals. They help but do not repair the case: error counters remain the worst-transferring family at a level-corrected R^2 of -1.05, against -2.39

raw. Each returns `nan` rather than raising on degenerate input, because constant series are the common case here rather than an edge case (§5.2). Implementations are validated against series whose answers are known by construction — a ramp, a sine, white noise, a random walk — in the accompanying test suite.

No single measure suffices, and the reason is instructive. Permutation entropy assigns 0.000 both to monotone counters, which have some 17850 distinct values, and to alarm states, which have two: a monotone series has exactly one ordering pattern and a step function almost never changes. Both are easy to forecast, for opposite structural reasons, and entropy cannot distinguish them (OI-11).

4.5 Forecast protocol

Skill is measured under rolling-origin evaluation with strict causality asserted per split. Errors are scaled by MASE with a denominator computed from the training window only; scaling by test-window statistics is the most common silent leak in forecasting benchmarks and would compress exactly the between-tier differences this study reads. The seasonal period is fixed per series *a priori* from its own cadence, with a non-seasonal robustness check (OI-5).

MASE is undefined where the in-sample seasonal-naive error vanishes. This is not an edge case here: a substantial share of control-plane series are constant (§5.1), and near-perfectly periodic windows yield denominators at the level of floating-point noise. Such series are reported as undefined rather than assigned a spuriously large score.

4.6 Pipeline and cost

The study is an eleven-stage pipeline, each stage writing a committed artefact so that any stage can be re-run without repeating the ones before it (Table 2). Bulk intermediates are sharded per partition and skipped when present, which makes an interrupted run resumable rather than repeatable.

Two engineering constraints shaped the design and are worth stating because they bound what the study could attempt. The archives hold roughly 55 billion rows against a machine with 62 GB of memory shared with other workloads, so extraction is selective and concurrency is derived from a memory budget rather than the core count. And every worker pins its BLAS and OpenMP pools to a single thread: with pools left at their defaults, five worker processes on a 24-core machine request 120 threads and the forecast sweep runs roughly 90 times slower, with no error and no warning to indicate it. The full sweep takes minutes once pinned.

4.7 Failure modes

Degenerate input is the common case here, not an edge case, so the pipeline is specified for it. A constant series yields an undefined MASE (the seasonal-naive denominator is zero) and is excluded from skill statistics and counted; so is a near-perfectly periodic one, whose denominator is at the level of floating-point noise and would otherwise produce an arbitrary ratio of two rounding errors. Every measure returns `nan` rather than raising on a series that is too short or has no variation. A series whose value column never parses as numeric is recorded as non-numeric and dropped. Where a forecaster fails to fit, the failure is recorded against that (series, model, origin) and the sweep continues, so one ill-conditioned fit cannot destroy a run.

4.8 Linking analysis

Skill is regressed on the predictability profile, cross-validated by holding out an *entire signal family*. A random split would let the model memorise family-specific quirks and report a flattering R^2; a held-out family forces the relation to transfer to signal physics it has never seen, which is what a portable difficulty law would require. Every R^2 is reported against a train-mean

Table 2: Pipeline stages and their committed outputs.

#	Stage	Output
1	inventory	partition catalogue (all 31 archives)
2	extract	selected partitions, disk-guarded
3	census	per-series descriptive table
4	taxonomy	a-priori labels + firing rule
5	sample	stratified series sample
6	features	21-measure profile per series
7	forecast	skill per (series, model, horizon, origin)
8	link	transfer, ablations, null control
9	figures	figures + backing CSV per figure
10	literature	screening table, bibliography
11	direct control	recursive vs direct forecasting

baseline, so "the measures explain skill" means better than knowing nothing rather than better than zero. Three ablations exist to attack the headline: dropping the instrument covariates, keeping only them, and removing the tie-biased entropy. Marginal correlations use a cluster bootstrap over metrics, because series of one metric across nodes are not independent (OI-9).

5 Results

5.1 The monitoring system is not stationary

Across the 31-month campaign we catalogue 10504 metric partitions comprising 573 distinct (plugin, metric) pairs. Only 104 — all of them IPMI — are present in every month. Ganglia's metric count runs 0, 33, 73, 49, 157, 177 and is not monotonic; SLURM's runs 20, 2, 0, 2, 36; Schneider first appears in 21-03 and Vertiv and Weather in 21-04.

Figure 1 shows the pattern. Any claim compared across months therefore confounds signal behaviour with changes to the instrument, and we restrict cross-month statements accordingly (OI-2).

5.2 Much of what is monitored never changes

Across 159506 series the fraction constant over a whole month varies by more than a factor of seven between families: 64.8% for Nagios, 47.0% Schneider, 42.9% Logics and Ganglia, 33.1% Vertiv, against 10.5% SLURM, 8.4% IPMI and 0% Weather. MASE is undefined for such series — the seasonal-naive denominator is zero — and they are excluded from skill statistics and counted rather than silently dropped. Figure 2 gives the breakdown.

Table 3: Protocol and model settings, fixed before results existed.

Setting	Value
Analysis grid	15 min, common to all families
Entity panel	96 primary entities, hashed
Horizons	1, 4, 12, 96 steps (15 min–24 h)
Rolling origins	4 per series
Training window	1500 most-recent points
Minimum series length	512 points
MASE denominators	same-step (primary), daily
Boosted trees	8/15, 24/60, 48/200 lags/iters
Neural	48–24 and 96–48 units, 40–60 iters
Ridge AR	order 24, $\lambda = 10^{-6}$
Seed	20230288

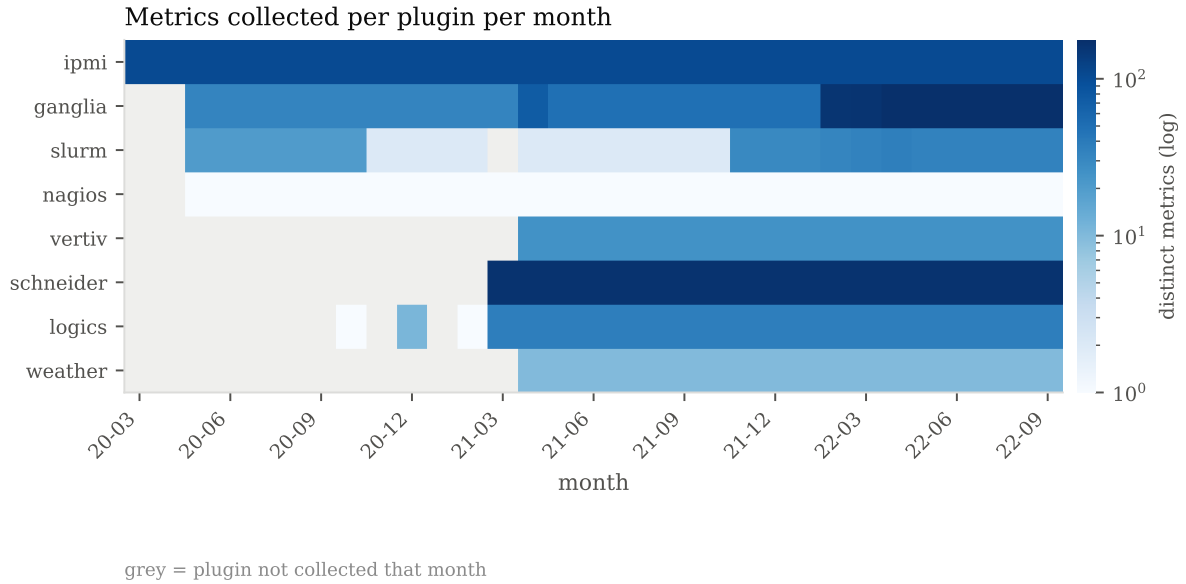


Figure 1: Metrics collected per plugin per month. Grey marks a plugin absent that month; the colour scale is logarithmic.

5.3 Regularity and interruption are different properties

Sampling irregularity is routinely summarised by the coefficient of variation of the inter-sample interval. On this data that measure is misleading: it reports IPMI as the least regular family, contradicting the dataset documentation. IPMI in fact has *zero* series with a jittery interval and 100% with a gap exceeding a hundred times the median — its sampling is exact and universally interrupted, and the rare enormous gaps dominate the coefficient of variation. Ganglia is the only family whose interval genuinely wanders (81% of series), which is what the documentation reports. We therefore measure jitter and interruption separately (OI-1); Figure 3 shows that the two are nearly orthogonal.

5.4 Forecast skill and model capacity

Figure 4 shows median skill by tier and horizon on 1082 series, pooling every model within a tier. Capacity does not buy skill: the naive tier leads at every horizon and the best single model throughout is persistence.

Pooling penalises a tier for containing a deliberately weak member, so we also compare each

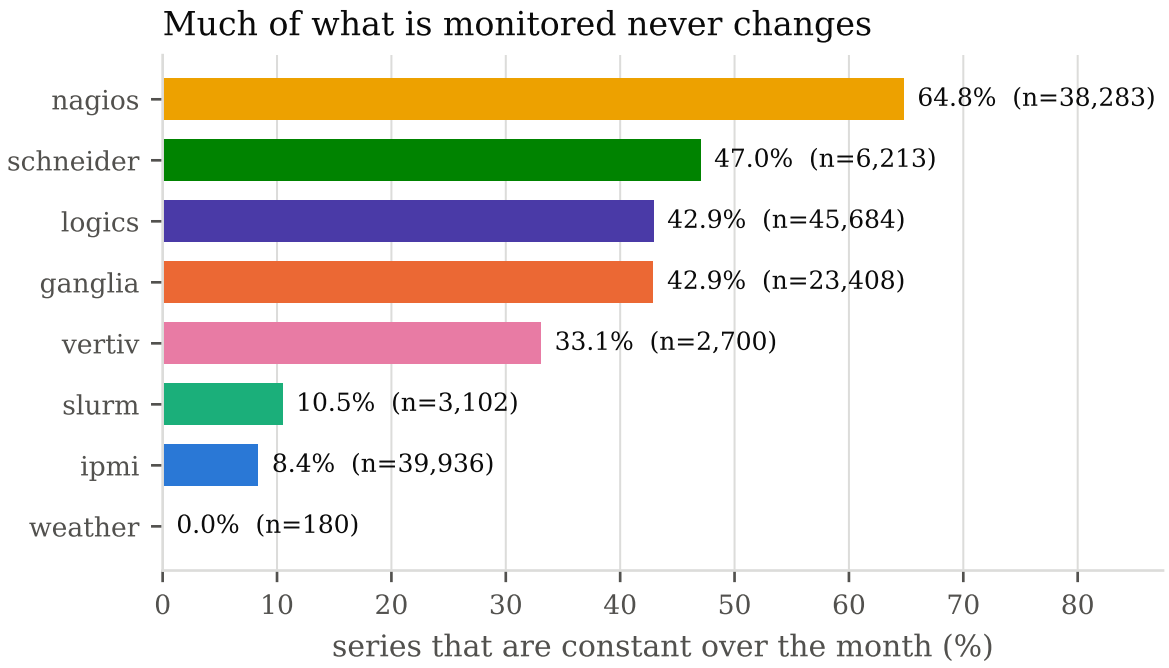


Figure 2: Share of series that never change over a month, by family.

Table 4: Median skill (1−MASE, same-step denominator) for the same learner forecast recursively and directly, against persistence, on 341 matched (series, origin) windows.

Horizon	Persistence	GBT recursive	GBT direct
15 min	0.767	0.320	0.392
1 h	0.499	-0.223	0.114
3 h	-0.023	-1.029	-0.259
24 h	-2.758	-3.692	-2.838

tier's *best* model per series, paired across series. That comparison is both more favourable to the sophisticated tiers and more conclusive against them. At a one-hour horizon the per-series best is 0.945 (IQR [0.70, 1.00]) for naive, 0.861 for statistical, 0.747 for boosted trees and 0.643 for neural networks; a paired Wilcoxon test of naive against each other tier rejects equality at every horizon and by a wide margin ($p < 10^{-20}$ against every tier at one hour, and $p < 10^{-90}$ against the neural tier). The ordering is not an artefact of pooling. At a 24-hour horizon even the best naive model is negative (-0.284), so the claim that no tier beats a same-step naive forecast holds at a day but not at three hours, where the per-series best naive model still scores 0.880. One qualification remains: the ordering depends on the baseline. Against a daily-seasonal denominator the statistical tier leads instead, and boosted trees overtake naive at the shortest horizon (OI-5).

5.5 Control: is this an artefact of recursive forecasting?

Every learned model above forecasts recursively — predict a step, append it, predict again — so error compounds, and the collapse of the learned tiers at long horizons might be that compounding rather than anything about the signals. This is the most plausible alternative explanation for the result, so we test it rather than caveat it. We fit the same gradient-boosted learner two ways on the same series, origins and horizons: recursively, and *directly*, trained to map a lag window onto the value h steps ahead in a single prediction.

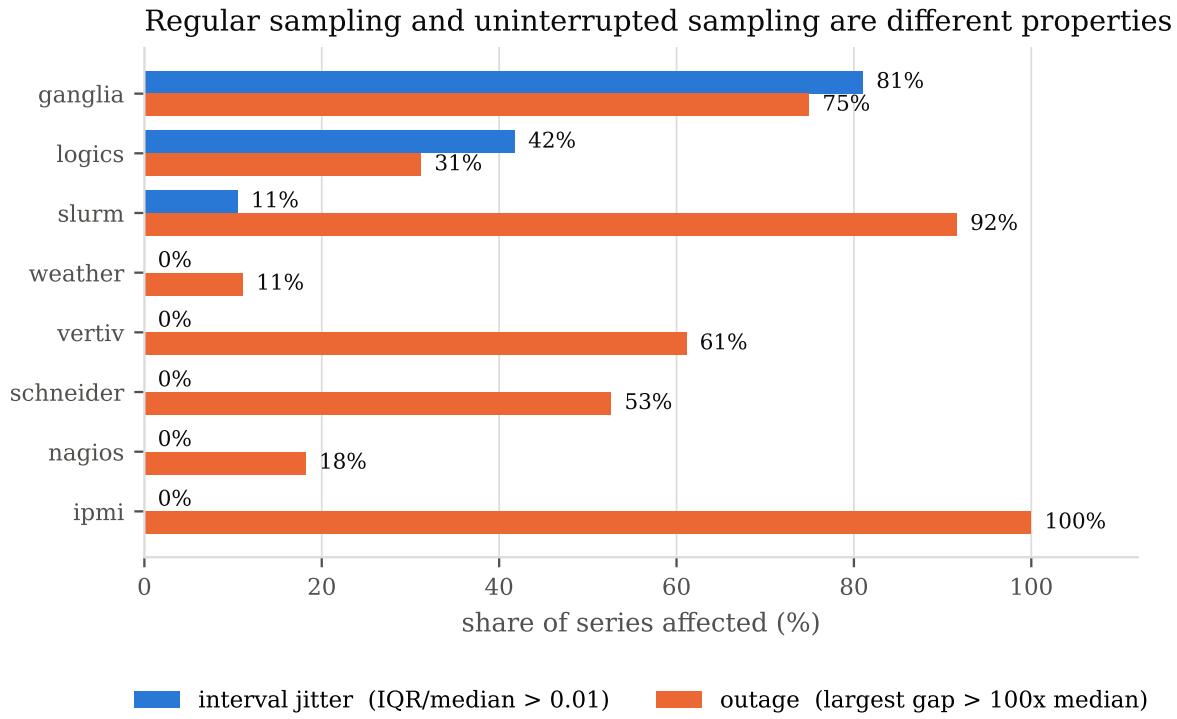


Figure 3: Interval jitter and outage incidence are different properties. IPMI has no jittery series and no uninterrupted ones.

Recursion does cost a great deal — forecasting directly recovers between 0.07 and 0.85 of skill depending on horizon, most at one and three hours — so the caveat was warranted. But the conclusion is unaffected: the direct model still loses to persistence at every horizon tested (Table 4). The capacity result is a property of the problem, not of the recursion.

5.6 Predictability does not transfer across families

Individually the measures correlate with skill. Cluster-robust over 261 metric clusters the strongest is the gzip compression ratio at $r = -0.42$ (95% CI $[-0.52, -0.31]$), ahead of repetition fraction $+0.37$ ($[0.26, 0.49]$), permutation entropy -0.33 and changepoint rate -0.30. That a general-purpose compressor outranks every hand-designed entropy measure is worth noting on its own.

Jointly, the raw picture looks poor: holding out each of 15 families in turn gives a mean out-of-sample R^2 of -0.19, a median of $+0.05$, positive for 9 of the 15, ranging from $+0.47$ to -2.39 (monotone error counters). Reported alone that number is misleading in two ways, and Table 5 shows why.

Ordering transfers; level does not. Within a held-out family the model ranks series correctly — median Spearman $+0.50$, significant at $p < 0.05$ for 9 of 15 families and reaching $+0.83$ — while placing that ranking at the wrong absolute level. R^2 penalises the level error; Spearman does not. Removing the level bias moves the mean R^2 from -0.19 to -0.01 and makes 10 of 15 families positive. The practical reading is that the profile can say which signals in an unfamiliar family will be harder than which, but not how hard any of them will be.

The failure is specific to family boundaries. Running the identical protocol with *random* groups of matched size gives a median R^2 of $+0.33$ across draws (2.5–97.5% range $[0.22, 0.38]$, mean $+0.32$) with every group positive, against -0.19 and 60% for the real families; over 100 draws no random grouping was as poor as the observed one ($p < 0.001$). Arbitrary grouping

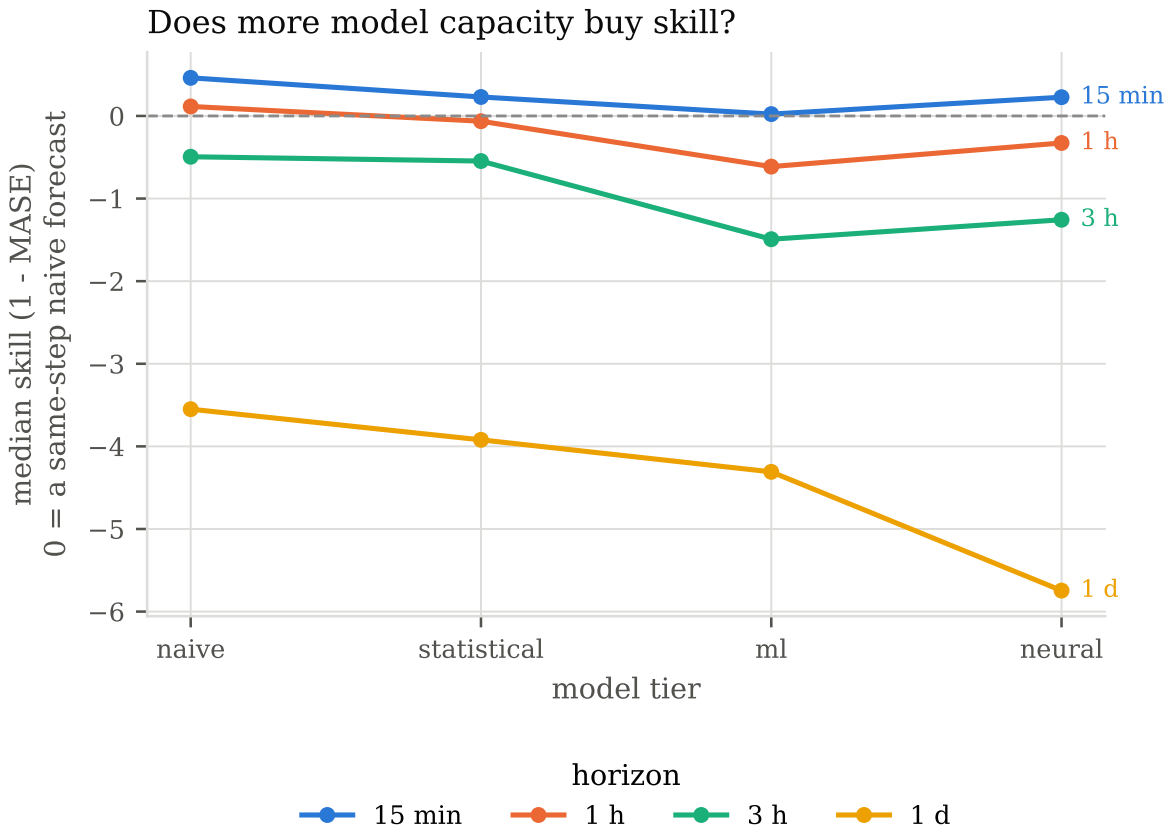


Figure 4: Median skill by model tier. Zero is a same-step naive forecast.

Table 5: Transfer to a held-out family, decomposed. Ordering transfers for most families; the absolute level does not, which is what the raw R^2 measures.

	mean	median	positive
raw R^2	-0.189	+0.053	9/15
level-corrected R^2	-0.011	+0.053	10/15
within-family Spearman	—	+0.495	9/15 at $p < 0.05$
random-group null (R^2)	+0.319	+0.329	15/15

therefore transfers easily, so the protocol is capable of transfer and the family axis carries real structure. Without this control the raw R^2 would be equally consistent with a protocol that simply cannot generalise at all.

The result is not an artefact of instrumentation. Using the instrument covariates alone collapses the model (-15.0 against -0.19 for the full profile), while removing them costs almost nothing (-0.21) — see Table 6. Whatever the measures capture, it is not how the data was sampled — which is the conditioning test OI-1 called for.

6 Threats to validity

The full register is maintained in `OPEN_ISSUES.md` and each item is cited from the text above. The load-bearing ones are:

oi-1 Ganglia's sampling irregularity confounds instrumentation with signal physics. Sampling regularity is therefore carried as a covariate and reported separately.

Table 6: Ablations. Removing the entropy family *improves* transfer; removing the instrument covariates costs almost nothing; using them alone destroys it.

Feature set	#	mean R^2
without entropy measures	22	-0.079
full profile	25	-0.189
signal measures only	21	-0.206
instrument covariates only	4	-14.999
predicting the global mean	—	-18.352

oi-2 The monitoring system changes over the campaign.

oi-6 No neural tier has been run; no claim of the form "not even deep models help" is licensed.

oi-7 The error floor is an estimate with a known bias direction, not an information-theoretic bound.

oi-11 Permutation entropy is biased downward by quantisation, and quantisation varies by family, so entropy is interpreted only alongside the measured quantisation step.

7 Conclusion

We set out to show that the difficulty of forecasting HPC telemetry is an intrinsic, measurable property of the signals. Half of that survives. Difficulty is not a modelling deficiency: across four tiers spanning naive baselines to neural networks, more capacity produced less skill, and beyond three hours nothing beat repeating the last value.

The other half holds in a weaker and more specific sense than we set out to show. The measures transfer to an unseen family in *ordering* — median within-family Spearman +0.40, significant for 10 of 15 families — but not in *level*, which is what drives the raw R^2 negative. A random-group null confirms the failure is about family boundaries specifically rather than about the protocol: arbitrary groups transfer easily ($R^2 = +0.33$, $p < 0.001$ against the observed value). So there is a portable component of forecast difficulty, and it is ordinal.

That distinction is the practical result. An operator facing an unfamiliar signal family can use a model-free profile to decide which of its signals to attempt first, and cannot use it to predict the accuracy they will get. Whether the level is learnable from a small labelled sample of the new family, which would make the ordinal result directly actionable, is the obvious next experiment and one this study does not run.

For operators the practical reading is narrower than we hoped but still useful: on this data, at these horizons, a persistence forecast is a strong default — significantly better than every heavier tier we tested, at every horizon — and the burden of proof lies with any claim that a larger model improves on it.

Disclosure

The author is a co-author of the M100 ExaData dataset paper [Borghesi et al., 2023] analysed here. No part of this study was performed as part of that data-collection campaign, and the analysis uses only the public release.

Data and code availability

All code, the versioned literature query set, the screening table, and every derived table are at `https://github.com/MSKazemi/hpc-signal-forecastability`. The underlying measurements are the public M100 ExaData release [Borghesi et al., 2023]. Results reproduce from the committed derived tables without downloading raw data.

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